

Builder Lens

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At the start of every new chat session where work will be done, ask for current date and time (SGT UTC+8), then ask: "Which project are we working on today?" before touching any code or design.

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1. Document Map

Sections 3 and 4 onwards are read together. General principles set the baseline. Project-specific observations add nuance and override where relevant.

Timestamp format on individual entries: Chat N - dd MMM yyyy SGT. Entries without a timestamp were present from initial document creation.

2. Prompt-Gate Questions

At the start of every new chat session where work will be done, ask these questions before touching any code or design:

- Step 1: "What is the current date, time, and timezone? (e.g. 27 Mar 2026, 06:48 SGT UTC+8)" - Added 27 Mar 2026 SGT
- Step 2: "Which project are we working on today?" - ensures the lens is applied precisely

General principles always apply. Project-specific context is opt-in.

3. General Principles

These apply to every project, regardless of tech stack, audience, or purpose. They are non-negotiable defaults.

3.1 Language and Copy

- UK English throughout - colour not color, organise not organize, centre not center
- Cambridge-Oxford grammar - no dangling modifiers, no careless fragments
- No em dashes or en dashes - replace with hyphens
- No self-promotional language - earned credibility and understatement only
- No consultant-style formatting in copy - no bold headers mid-sentence
- Plain, humble, direct - closer to speaking to a friend than writing a brochure
- Short hashtag lists over volume - 4 to 6 maximum
- No emojis unless explicitly approved for a specific context

3.2 UI Copy and Tone

- Third person neutral - the tool shows, it does not tell
- Never prescribe - show the landscape, respect the user's judgement
- Indicative framing - starting point, not verdict
- Warm error messages - the system speaks like a thoughtful colleague
- Remove timing claims unless they can be guaranteed
- Labels as invitations - Fresh Graduate not You as a fresh graduate
- Remove redundant instructions - every word must earn its place
- Plain language for AI concepts - no jargon for someone who has never used AI automation before - Chat 6 - 18 Mar 2026 SGT
- Error messages must be proportionate - Showing the closest matches found is honest and calm - Chat 6 - 21 Mar 2026 SGT

3.3 Visual Design

- Clean and restrained - white surfaces, subtle borders, generous whitespace
- Colour encodes meaning, never decoration
- Mobile first - 375px before 1440px
- Progressive disclosure - show minimum, reveal on demand
- Consistent font sizes across equivalent content types
- Hover states consistent - if one card has hover, all cards of same type must
- Buttons always visible - never in same flex row as wrapping badge elements - Chat 6 - 18 Mar 2026 SGT
- Help tips belong directly adjacent to the thing they explain, not at the bottom - Chat 6 - 21 Mar 2026 SGT

3.4 Product Decisions

- Say do not build yet - think before committing to code
- Ship working before adding complexity - minimum viable first
- Each surface does a different job - no two panels repeat the same content
- Cut features that serve the builder's ego rather than the user's need
- Separate advisory conversations from code-change conversations
- Floating action or persistent nudge required when new content appears off-screen
- Onboarding must not interrupt. Inline blink on the first relevant element is less disruptive than an overlay - Chat 6 - 21 Mar 2026 SGT

3.5 Technical Standards

- Single source of truth for constants - LEVELS, C object, TECH_META
- grep for critical globals before packaging - C, LEVELS, claudeCall, PERSONA_CONFIG
- Successful build does not mean working app - browser is the honest source of truth
- React inline styles do not support CSS functions - use className
- Version comment on line 1 of App.jsx on every change
- pickerFullError and similar state variables must be verified as props at call site after refactoring
- For API calls that models handle unreliably, use client-side hardcoded lookup before calling the API - Chat 6 - 21 Mar 2026 SGT

- Child component props that are used inside the component but not explicitly passed at all call sites must have default prop values - Audit session - 22 Mar 2026 SGT
- Vite v5 esbuild rejects multi-line async arrow functions as JSX props. Always extract to named functions above return statement - v2 24 Mar 2026 SGT

3.6 Review Behaviour - What to Expect

- Will question decisions before they become code - this is not obstruction
- Will flag when a fix introduces new risk even if the fix itself is correct
- Will push back on copy that is vague, inflated, or hedged
- Will ask for the simplest version first
- Will state analysis and ask for confirmation on ambiguous scope

3.7 Working Style Preferences

- Iterative - one fix at a time, verify before next
- Screenshot-first for UX issues - visual evidence over text description
- Constraint-based prompting for build sessions - explicit scope prevents drift
- Honest about what is not working - does not chase false comfort
- Platinum Compact Framework as response standard - structured header, no prose, download links only
- Serialisation line at top of every response: No NNN - DD-MM-YY HH:MM | SN-NNN - wfg-plans Session 5 - 27 Mar 2026 SGT
- Ask Lead for confirmed date and time at session open before any work - wfg-plans Session 5 - 27 Mar 2026 SGT

4. Project A - takearoundabout.com (AI Skills Analyser)

AI Skills Analyser. Built from 13 Mar 2026. Live at takearoundabout.com.

4.1 UI Copy Decisions Observed

- found for replaced with similar to - found asserts completeness the tool cannot guarantee - Chat 6
- Your next move replaced with What to do Next - second person removed - Chat 6
- Kickstart replaced with Apply - warm, neutral, no overlap with Prompt label - Chat 6
- Generating sophisticated prompt replaced with patient, humble waiting copy - Chat 6
- Copy and Copy prompt replaced with Copy Prompt and Copy Instructions with Copied. Ready to Paste feedback - Chat 6
- Could not load the full list replaced with Showing the closest matches found. Search again if you need more results. - Chat 6 - 21 Mar 2026 SGT
- Skill Detail renamed Skill Analysis - aligns with heading inside the tab - Chat 6 - 21 Mar 2026 SGT
- NEXT PHASE all-caps label renamed to What to do next - uppercase technical labels are not user language - Chat 6 - 18 Mar 2026 SGT
- Renamed Copy button to Copy prompt - Copy was ambiguous alongside the Copy and go readiness badge - Chat 6 - 18 Mar 2026 SGT
- Renamed Open in new tab to Explore similar role - Open in new tab sounds like a browser action - Chat 6 - 18 Mar 2026 SGT

4.2 Design Decisions Observed

- Left border on What to do Next card removed - white card cleaner than coloured border - Chat 6

- Navigation section with 2px accent border - visual landmark for tab discovery - Chat 6
- First-time loading explanation - two cards below spinner, shown once only - Chat 6
- 3-second left border highlight on target skill after jump - Chat 6
- CoachMark overlay removed - inline blink on first AI skill is less disruptive - Chat 6 - 21 Mar 2026 SGT
- Help tip inside {isOpen} block - hides with panel on collapse, visible on expand - Chat 6 - 21 Mar 2026 SGT
- Blink duration 20 seconds - sufficient dwell time for first-time users on mobile - Chat 6 - 21 Mar 2026 SGT
- Copy prompt button moved below prompt text - button was invisible on narrow screens when badges wrapped - Chat 6 - 18 Mar 2026 SGT
- Readiness badges use flat label style (borderRadius 4) not pill - they are status labels, not actions - Chat 6 - 18 Mar 2026 SGT
- Floating nudge button for below-fold content - fixed-position bottom button appears when more results load below fold - Chat 6 - 18 Mar 2026 SGT
- Coherence notice in results header - amber when ESCO skills may not match role. Refresh skills with AI + Search again buttons - v2 24 Mar 2026 SGT
- Function keyword notice in picker - blue info notice when bare discipline name searched. Suggests specific titles. Proceed anyway button - v2 24 Mar 2026 SGT
- May not apply amber pill on skill row - Sonnet scored skill as low-relevance. Hover tooltip explains - v2 24 Mar 2026 SGT

4.3 Structural Decisions Observed

- pickerFullError must be passed as prop to OccupationPicker - learned from v1.8.2 ReferenceError - Chat 6
- setJumpToSkill delayed 80ms after setActiveTab - SkillGroupedView must mount first - Chat 6
- Haiku timeout 90s for maxTokens > 2500 - 55s too tight for 40-50 result searches - Chat 6
- Background picker: 3-retry async loop [0ms, 2000ms, 4000ms] before giving up - Chat 6 - 21 Mar 2026 SGT
- SENIOR_MGMT_LOOKUP: hardcoded entries intercept doSearch() before any API call. Now includes fuzzy substring fallback - Chat 6 - 21 Mar 2026 SGT + v2 24 Mar 2026
- firstBlinkSkill state replaces coachMark + coachSkillName. Targets first non-Human-Led skill. Clears via setTimeout after 20s - Chat 6 - 21 Mar 2026 SGT
- analysisCancelRef pattern: doAnalyse increments cancel counter at entry. Every setState guards with cancelId check - Audit session - 22 Mar 2026 SGT
- validateJobTitleInput() gate: fires before any API call in doSearch() and URL param handler - Audit session - 22 Mar 2026 SGT
- doAnalyse lookup intercept: lookupSeniorMgmt fires on occ.title at analysis time - v2 24 Mar 2026 SGT
- forceHybrid flag: doAnalyse(occ, { forceHybrid: true }) skips ESCO and uses Claude getSkills() directly - v2 24 Mar 2026 SGT

4.4 Micro-Managing Patterns Observed

Copy triggers

- Any sentence containing genuinely, honestly, or straightforward - always removed
- Brochure language - phrases like empowering users, transforming careers, cutting-edge
- Over-qualified disclaimers that repeat what the design already shows
- Hashtag lists longer than 6 - always cut
- LinkedIn copy that sounds like an announcement rather than a conversation
- Instructional hints that explain what a visual affordance already communicates - Chat 3
- Caution notices that say some when they could name the specific items - Chat 3
- Prompt examples that use generic context when Singapore-specific context is available - Chat 5

- Technical tier names used as user-facing copy - MEDIUM, automated workflow must never appear in text for end users - Chat 6 - 18 Mar 2026 SGT
- Next phase text that begins with a lowercase letter - strip the prefix and capitalise - Chat 6 - 18 Mar 2026 SGT

UI triggers

- Percentage numbers next to visual elements that already encode the information
- Hover-only interactions on any element that carries meaning - must be always visible
- Two panels on the same screen saying the same thing in different words
- Error states that include promotional elements - no LinkedIn button in error messages
- Labels that prescribe rather than invite
- Horizontal scrolling navigation on mobile - stacked cards preferred - Chat 3
- Vertical misalignment between icon or bubble and adjacent text - Chat 3
- Subtitles that restate the heading - Chat 3
- Picker grouping that places Apprentice, Intern, Trainee under a single sector - these are cross-industry roles - Chat 6
- Copy prompt button not visible - if action button shares row with wrapping badges it disappears on narrow screens - Chat 6
- Picker showing complete count before loading is complete - false precision - Chat 6

Technical triggers

- Preview JSX using wrong model string - Haiku string does not work in artifact sandbox
- Sanitiser in proxy breaking valid prompts - never sanitise on the server side
- Proxy rewrite pattern intercepting serverless functions - must use negative lookahead
- AI tool rated NA but level showing AI-Assisted - logical contradiction caught from screenshot
- CSS functions in React inline style props - clamp(), calc(), min(), max() fail silently - Chat 3
- Module-level JSX constants that reference undefined variables - crashes at load - Chat 3
- Missing module-level constants after chat compaction - grep before packaging - Chat 3
- Stale prop references when copying component structure - Chat 5
- SYSTEM_RATE token ceiling: adding instructions to a system prompt close to context limit causes silent failures - Chat 6
- pickerCancelRef pattern: any async call writing shared state needs a cancellation flag - Chat 6
- Prompt diversity requires explicit enforcement: model defaults to 3-4 safe techniques without DIVERSITY RULE - Chat 6
- Multi-line async arrow functions as JSX props rejected by Vite v5 esbuild - v2 24 Mar 2026
- sed font-size cascade: running sed s/12/10/ then s/10/9/ corrupts all original 12s - always verify before and after - v2 24 Mar 2026

4.5 What He Never Compromises On

- Honesty in copy - no false precision, no completeness claims the tool cannot support
- Accessibility - semantic HTML, ARIA labels, skip nav, keyboard navigation
- Mobile experience - tested on iPhone, not just desktop resize
- ESCO integrity - no invented titles, no sector misclassification
- Token budget discipline - raise when needed, justify the increase
- Build trail integrity - version on line 1, verify before package, no assumptions from memory
- Source of truth - ESCO as traceable public data, ISCO-08 attributed correctly, no invented data
- Free to users - monetising felt wrong given the underlying data is free
- No personal data collected - transparency about data practices is non-negotiable
- Human dignity in the tool's tone - especially for roles showing high automation exposure

- User control over AI substitution - when AI selects a role, user must be shown what happened and given a choice - Chat 6

5. Project B - wfg-plans (NTUC-ARU Workforce Planning Dashboard)

wfg-plans v0.1.5-beta. Built from 23 Mar 2026. Internal tool - Confidential. Updated 27 Mar 2026 SGT.

5.1 Non-Negotiable Constraints

- Canonical hierarchy is absolute: Group > Division > Department > Section > Estab Title. Every tab, every table, every label must reflect it. Any drift from this is a data integrity failure, not a UX choice.
- All 1,521 RAW_ESTAB rows must be visible at every drill level. A post assigned directly to a division or department with no section must appear in the tables. Hiding it is a data accuracy failure.
- Grade keys are always zero-padded: N09, N08, N07 etc. Using unpadded keys causes silent table failures.
- Always filter RAW_ESTAB by g_raw or GRP_RAW[groupId] - never by GROUPS.name. T&T vs TRAINING & TRANSFORMATION mismatch returns empty results.
- WFG-Plans beta.zip is the fixed zip filename. Never serialise with version or date. The Lead runs terminal deploy commands and must not retype filenames between sessions.

5.2 Build Trail Standards - wfg-plans specific

- esbuild Unterminated regular expression error is a JSX structural parse failure. Cause: unclosed map callback nesting. Trace brace depth from enclosing Card to error line. - Session 4
- Non-ASCII characters in JSX string literals cause esbuild parse failures. Replace all with ASCII equivalents before packaging. - Session 4
- Vercel MIME error is a downstream symptom of a failed build. Diagnose npm run build first, not vercel.json routing config. Three attempts at vercel.json were wasted because the root cause was in App.jsx. - Session 4
- React array returns from render functions: returning an array of tr elements from a function called inside tbody renders correctly without a Fragment wrapper. - Session 4
- Absolute-positioned dropdown menus are clipped by overflow:hidden on ancestor containers. - Session 4
- iCloud Drive folders accessed from MacBook terminal require the full path: ~/Library/Mobile Documents/com~apple~CloudDocs/... - Session 5
- GitHub Desktop cannot see a folder as a repository unless .git exists inside that specific folder. git init must be run inside the project folder, not a parent. - Session 5
- GitHub email privacy setting blocks terminal pushes. Use noreply address in git config or temporarily disable the setting. - Session 5
- statsGroupedBySection must handle blank-section posts by rolling them into a dept-level row. Without this, Director posts assigned at dept level are invisible in section drill. - Session 4
- tblDept and tblSec are independent table drill states. Section click in left tree must set tblDept(dept name) before setTblSec. Missing tblDept causes tables to stay at All Departments level. - Session 5
- tblSec must reset on selDept change (search clear, dept list click, clear button). Missing reset causes tables to show previous dept section data in new dept context. - Session 5

5.3 UX and Data Philosophy

- The pivot table pattern is the right one. Tables are both display and navigation. Clicking any row drives the entire tab drill state, same as clicking the chart or tree above. - Session 4
- User never needs to scroll up to change the data view. Tables and scroll-to-tables handle this. - Session 4
- Data accuracy as baseline, not feature. When a post is assigned directly to a division or department with no section, it must appear in the tables. Not showing it is a data accuracy failure, not a design choice. - Session 4

- The five-scope comparison panel in Workforce Statistics is self-contained. Scope selections do not affect other tabs. - Session 3
- Hierarchy labels on all tables: Estab by: Group / Division / Department / Sub-Dept. These change dynamically based on drill level. - Session 3

5.4 Micro-Managing Patterns Observed - wfg-plans

Data triggers

- Grade count table showing zero rows - silent failure from missing `g: {}` in `statsFromEstabRows` output
- `WorkforceStatsTable` showing wrong scope - `tblDept` or `tblSec` not set or reset correctly
- Tables not refreshing after tree click - missing `onRowClick` prop or missing `setTblDept` call in section click handler
- Division chart bar click not clearing child state when clicked again - `deselect` logic missing
- `statsGroupedBySection` grouping unsectioned posts into (Other) instead of dept-level row - wrong grouping logic

Technical triggers

- Babel-banned patterns in `wfg-plans` code: `?? / ?.` / module-level IIFE / direct `React.createElement()` calls
- Non-ASCII characters (em dash, triangle, dagger) in JSX string literals - `esbuild` parse failures
- `manualChunks` in `vite.config.js` - causes cross-chunk React crash
- Vercel MIME error - always check `npm run build` locally first before any `vercel.json` changes
- ZIP file named anything other than `WFG-Plans beta.zip` - breaks Lead terminal workflow

6. Future Projects

- Project C - to be added
- Project D - to be added

7. Instructions for Future Chat Sessions

At the start of every session

- Ask Lead for current date, time, and timezone before any work: "What is the current date, time, and timezone?" - Added 27 Mar 2026 SGT
- Upload this document alongside Builder Framework, Builder Persona, and handover document
- Ask the prompt-gate questions from Section 2 before doing any work
- Confirm build file state: `head -1 App.jsx` and `wc -l App.jsx`
- Do not ask for confirmation of preferences already documented here - act on them

During the session

- Apply Section 3 general principles silently - do not announce you are applying them
- Anticipate the micro-managing triggers in Section 4 and 5 before presenting output
- If a decision conflicts with a documented preference, flag it explicitly and explain why

At the end of the session

- If new micro-managing patterns or preferences were observed, surface them
- Ask: shall I add these observations to the Builder Lens before we close?
- Use file naming convention: `builder-lens-(dd-mm-yy hhmm).docx`

Session 6 Additions - Added 28-3-26 1201 SGT

3.5 Technical Standards - wfg-plans Session 6 addition

- sed replacements with plural/singular pairs: always execute plural first. Running UNION before UNIONS double-hits UNIONS. - wfg-plans Session 6 - 28 Mar 2026 SGT

5. Project B - wfg-plans - Session 6 additions

Added 28-3-26 1201 SGT

5.2 Build Trail Standards additions

- sed redaction: plural form before singular form prevents double-hit. UNIONS -> FORCES before UNION -> FORCE. Residual count must be verified at zero for all terms before packaging. - Session 6 - 28 Mar 2026 SGT
- index.html <title> is a separate file from App.jsx. Redaction passes on App.jsx do not cover index.html. Both files must be checked after any redaction. - Session 6 - 28 Mar 2026 SGT
- README.md version and title must be updated alongside App.jsx and index.html at every version bump. - Session 6 - 28 Mar 2026 SGT
- git push via terminal is more reliable than GitHub Desktop when files have been replaced by drag-and-drop. Use: git add . && git commit -m 'message' && git push - Session 6 - 28 Mar 2026 SGT

5.3 Redaction Decisions Observed - v0.2.0-beta

- Version identity change (not a patch): replacing organisation names throughout a data-embedded dashboard warrants a version bump (v0.1.5-beta -> v0.2.0-beta). - Session 6 - 28 Mar 2026 SGT
- Side effects must be identified before executing: group selector pill labels, WorkforceStatsTable headers, funding donut labels, footer lines, window title, README.md. - Session 6 - 28 Mar 2026 SGT
- Seatrrium Union (mixed case 'Union') correctly excluded from FORCE replacement. Side effect analysis must distinguish case variants. - Session 6 - 28 Mar 2026 SGT
- Redaction map confirmed: NTUC->OUVD (21), EAC->FBD (49), Workforce->People (68), OTC-LLI->PUDMM (3), TADM->PCWCR (2), UNION/UNIONS->FORCE/FORCES (24). - Session 6 - 28 Mar 2026 SGT

5.5 Micro-Managing Patterns - Technical triggers additions

- index.html title not updated after App.jsx redaction - window tab shows original org name on live URL - Session 6
- README.md version still at old number after version bump - Session 6

- Redaction residuals not verified at zero before packaging - causes sensitive term leakage - Session 6